ALEJANDRO LUIS ARMAS BRAITHWAITE

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EDUCATION

University of Chicago — Chicago, IL
M.S. in Computational Analysis and Public Policy — GPA: 3.59/4.0 — **August 2026**
Relevant Coursework: *Machine Learning, Advanced Statistics, Algorithms, GIS & Spatial Analysis*

Pacífico Business School — Lima, Peru
M.A. in Economics — **December 2019**

Universidad del Pacífico — Lima, Peru
B.A. in Economics — Honors: Top Third of Students in 2015 and 2016 — **December 2017**

RESEARCH PROJECTS

Detecting Political Ideology in News Media Articles — Chicago, IL
Research Project, in collaboration with a faculty researcher — **Summer 2025**

- Collected ~60,000 news articles and images (~10K per outlet) from 6 news media outlets using APIs, Selenium, and Requests.
- Cleaned and structured the multimodal (text + image) dataset for downstream NLP and computer-vision classification of political bias.

Chicago Livability Score Project — Chicago, IL
University of Chicago — **Winter 2025**

- Built crime, school, and environment indices for 60 Chicago ZIP codes from 257K crime records and 94K pollution complaints.
- Mapped point-level records to ZIP codes via spatial joins with Census shapefiles (GeoPandas, Shapely) and wrote unit tests (pytest) for a 4-person team's interactive Flask/Folium livability map.

Predicting Dangerous Streets with ML — Chicago, IL
University of Chicago — **Spring 2025**

- Built and compared Logistic Regression, Decision Tree, Random Forest, and Neural Network (MLP) classifiers to predict high-risk NYC street segments, using SMOTE to address class imbalance.
- Best model (tuned Random Forest) reached a 0.59 test F1 score vs. 0.36 for a random baseline and 0.49 for logistic regression, capturing non-linear effects of lane count, speed limits, and traffic calming.

PROFESSIONAL EXPERIENCE

The University of Chicago – Policy Lab (with IOM) — Chicago, IL
Policy Lab Consultant | Client: IOM — **August 2025 – December 2025**

- Co-developed (5-person team) a standardized, Sendai-aligned cost model estimating ~US$395M in annual investment needs for gender- and migration-inclusive Disaster Risk Management (DRM) across 5 South American countries.
- Adjusted GDP benchmarks with 4 normalized IDB governance and vulnerability indices, allocated costs across 5 policy dimensions, and validated findings with IOM and UN country offices.

The University of Chicago – Harris School of Public Policy — Chicago, IL
Teaching Assistant – Machine Learning for Public Policy — **August 2025 – December 2025**

- Led TA sessions, held office hours, and graded assignments and quizzes for a section of ~30 graduate students.
- Developed reproducible Python scripts and datasets implementing machine learning methods for public policy applications, supporting model replication and empirical analysis.

Universidad del Pacífico – Faculty of Economics and Finance — Lima, Peru
Lecturer (3 years) | Teaching Assistant (7 years) — **March 2014 – July 2024**

- Taught applied economics and econometrics courses for 10 years, guiding students through empirical analysis with real-world datasets.

Ministry of Foreign Trade and Tourism (Peru) – COPESCO — Lima, Peru
Public Investment Formulation Assistant — **November 2023 – April 2024**

- Monitored evaluation and compliance for 10+ public tourism infrastructure projects, assessing proposals under Peru's Invierte.pe system.
- Cleaned and structured administrative datasets for feasibility analysis and descriptive statistical reporting.

Ministry of Culture (Peru) – Economics Studies Unit — Lima, Peru
Consultant — **October 2020 – February 2021**

- Consolidated 5 quantitative data sources and qualitative inputs into a centralized Excel dashboard for budget and policy analysis.
- Produced analytical tables and visualizations to support budget allocation decisions and policy briefs.

TECHNICAL SKILLS

Programming Languages: Python (pandas, scikit-learn, GeoPandas, Shapely, Selenium, Altair), R, Stata, SQL, JavaScript (D3.js)

Software & Tools: Git, Jupyter Notebooks, pytest, Excel (Pivot Tables, VBA, dashboards); **Languages:** English (Fluent), Spanish (Native)

Research Methods: Statistical modeling, machine learning, web scraping & API data collection, data cleaning & integration, data visualization, spatial analysis, reproducible research.